

Mashiat Amin Farin

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EDUCATION

University of Texas at Dallas (PhD in Software Engineering)
University of Dhaka (BS in Software Engineering)

Expected graduation Fall 2030

CONFERENCES

M A Farin, B Thuraisingham, "A Four-Axis Systematization of AI and Cybersecurity in the Era of Large Language Models", IEEE BigDataSecurity 2026.

M A Farin, S Hassan, J Arroyo, "Mapping Decentralized Autonomous Organization Governance Across Chains: An Updated, Multi-Platform Dataset", MSR 2026.

M A Farin, S Sikder, A Y Raad, M T Ahmed, T Tahlil, "HealthChain: A Blockchain-Based Framework for Electronic Health Record Management System", WETSEB 2025.

S Sikder, **M A Farin,** M A Islam, T Tahlil, M T Ahmed, "Unlocking DeFi Literacy: Understanding NFT Market Microstructure in the Decentralized Finance Landscape", ICBC 2024.

RESEARCH EXPERIENCE

University of Texas at Dallas | Research Assistant **Jan 2026 - Present**

- o Investigating software engineering challenges in the development, evaluation, and governance of AI-driven software systems.
- o Studying trustworthy integration of large language models into software engineering workflows and applications.
- o Conducting empirical analyses of emerging AI-enabled software systems to identify reliability, security, and challenges.

Universidad Complutense de Madrid | Research Assistant **Apr 2025 - Present**

- o Empirically studied DAOs and governance mechanisms across blockchain ecosystems.
- o Constructed and analyzed a large-scale multi-platform DAO governance dataset, contributing to a publication at MSR 2026.
- o Investigating egalitarianism and representation in decentralized governance via analysis of participation and voting dynamics.

Data & Design Lab, CSE, University of Dhaka | Research intern **Nov 2023 - Feb 2025**

- o Analyzed electrical load and feeder datasets from Dhaka city substations to evaluate operational performance and quality.
- o Developed and evaluated machine learning approaches for seasonal load forecasting and substation performance assessment.
- o Compared forecasting methodologies and communicated findings through technical reports and visualizations.

IIT, University of Dhaka | Research Associate **Jan 2023**

- o Contributed to research on machine-learning-based cyberattack prediction and threat intelligence systems.
- o Conducted experimental evaluations of machine learning models and analyzed performance across multiple datasets.
- o Presented research findings through workshops and technical demonstrations.

IIT, University of Dhaka | Undergraduate Student Researcher – Computer vision under Dr. Mohammad Shoyaib **Aug – Dec 2022**

- o Surveyed activity recognition and computer vision techniques for industrial applications.
- o Analyzed contemporary research directions and methodological trends in computer vision systems.

IIT, University of Dhaka | Undergraduate Student Researcher – NLP under Dr. Ahmedul Kabir **May 2022 – May 2023**

- o Investigated Bangla language resources including word embeddings, stemming techniques, and hate speech datasets.
- o Conducted literature reviews and exploratory analyses of NLP methodologies for low-resource languages.

SELECTED ACADEMIC PROGRAM

Summer School in the Security and Governance of Collective Digital Infrastructures (SGCDI 2026) **Jul 2026**
Max Planck Institute for Security and Privacy, Germany | July 2026

Selected participant in an interdisciplinary summer school bringing together researchers in security, privacy, governance, and socio-technical systems. Participated in lectures, discussions, and a collaborative research hackathon on collective digital infrastructures.

RESEARCH INTERESTS

Software Engineering for AI and LLM-Based Systems, Human Factors in Software Engineering, Empirical Software Engineering, Trustworthy AI Systems, Governance and Socio-Technical Software Ecosystems

AWARDS AND HONORS

2026 SIGSOFT CAPS Award for MSR 2026

2026 8th in The 2026 Spring IEEE International Contest on Software Testing

2024 Finalist at International Blockchain Olympiad 2024

2024 Honorable mention at National Blockchain Olympiad 2024

2022 Bronze medal at SDG:3 Category in International Blockchain Olympiad

PROFESSIONAL EXPERIENCE

Brain Station 23 | *Associate Software Engineer*

Jan - Nov 2024

- o Evaluated smart contract optimization techniques for reducing transaction costs and improving execution efficiency.
- o Investigated software assurance and auditing approaches for identifying vulnerabilities in blockchain applications.
- o Researched blockchain data indexing and retrieval techniques using GraphQL and decentralized indexing frameworks.

Grameen Communications | *Blockchain R&D intern*

Jun-Jul 2023

- o Researched the existing microcredit system to assess blockchain's potential for enhancing security.
- o Developed a blockchain prototype for the current system, formulated & implemented smart contracts to enhance user service.

SELECTED PROJECTS

FinCube | Solidity, Hardhat, The Graph, NestJS, React

Mar 2024

- o Designed a decentralized governance model for cross-border currency exchange, addressing limitations of existing services.
- o Developed and evaluated smart-contract-based mechanisms for currency swaps, governance voting, and business registration.
- o Implemented blockchain indexing infrastructure to support interoperability between decentralized and traditional systems.

DriveChain | Solidity, Hyperledger Firefly, React

Jun 2023

- o Designed a blockchain-based architecture for transportation licensing, registration, and taxation workflows.
- o Developed smart-contract-based automation mechanisms and document authentication for digital transportation records.
- o Explored NFT-based representations of vehicle ownership and registration assets.

HealthChain | Solidity, Hyperledger Firefly, React

Mar 2022

- o Led a four-member team in designing a distributed healthcare record system for large-scale patient record management.
- o Conducted an extensive review of healthcare infrastructure challenges and blockchain-based healthcare solutions.
- o Developed a decentralized framework for secure and auditable electronic health record management.

Hate speech analysis from social media comments | Pandas, NLTK, NumPy, Google Colaboratory, Python

Jul 2022

- o Evaluated machine learning models for Bangla hate speech classification using a corpus of 50,000 social media comments.
- o Compared multiple learning approaches including Logistic Regression and SVM using TF-IDF feature representations.
- o Achieved 89% classification accuracy on the evaluation dataset.

ADDITIONAL PROJECTS: MiniChessAI • WumpusAI • Fire Prevention System • Decentralized Microfinance System

DEVELOPMENT & PROBLEM SOLVING SKILL SET

Technical Skills:

Python, Java, C++, R, SQL, Solidity

Software Engineering & Data Science:

Pandas, NumPy, Scikit-Learn, GraphQL, Jupyter, Docker, Git

Web & Distributed Systems:

React, Node.js, Flask, Hyperledger Firefly, Hyperledger Besu, PostgreSQL, MongoDB

LEADERSHIP & SERVICE

IEEE Computer Science Student Branch University of Dhaka | *General Secretary*

Jan 2023 – Jan 2024

IIT Software Engineering Community | *Vice President*

Jan 2023 – Jan 2024